

Textbook-style Student Notes
TYBCA – BCA30040 INTRODUCTION TO CLOUD COMPUTING

UNIT II – CLOUD SERVICE MODELS

Programm/Class: TYBCA

Primary Reference: Thomas Erl, Ricardo Puttini & Zaigham Mahmood, *Cloud Computing: Concepts, Technology & Architecture*.

Duration: 6 Lectures

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UNIT-II: Cloud Service Models

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1. Introduction to Cloud Service Models

1.1 Cloud Service / Delivery Models

1.1.1 Meaning of Cloud Delivery Model

A **cloud delivery model** describes how cloud-based IT resources are packaged and made available by a cloud provider to a cloud consumer.

In the reference book, cloud delivery models are used to describe different ways in which cloud-based IT resources can be provided and consumed.

The three established delivery models discussed in the book are:

1. **Infrastructure-as-a-Service (IaaS)**
2. **Platform-as-a-Service (PaaS)**
3. **Software-as-a-Service (SaaS)**

The book's index specifically identifies these three models under **Cloud Delivery Models**.

Simple understanding

IaaS → Infrastructure

PaaS → Platform

SaaS → Software

1.1.2 Purpose of Cloud Delivery Models

Cloud delivery models help identify:

- What type of IT resources are being provided
- What the cloud provider manages
- What the cloud consumer manages

- How much control the consumer has
- What activities the consumer can perform

Therefore, the delivery model establishes an important relationship between the **cloud provider** and **cloud consumer**.

1.1.3 Cloud Provider

A **cloud provider** is the party that provides cloud-based IT resources and services to consumers.

Depending on the selected delivery model, the provider may manage:

- Physical infrastructure
- Virtualized infrastructure
- Platforms
- Software
- Supporting resources

The book separately discusses cloud delivery models from both the **cloud provider perspective** and the **cloud consumer perspective**.

1.1.4 Cloud Consumer

A **cloud consumer** is the party that consumes cloud-based IT resources or services.

The activities of the consumer depend on the delivery model.

For example:

- In **IaaS**, the consumer configures infrastructure and manages required software.
 - In **PaaS**, the consumer develops, tests and deploys applications/services.
 - In **SaaS**, the consumer mainly uses and configures the cloud service.
-

1.1.5 Level of Consumer Control

The delivery models differ in the amount of control provided to the cloud consumer.

MORE CONSUMER CONTROL

|
IaaS
|
PaaS
|

SaaS

|

LESS CONSUMER CONTROL

Among the three base models:

IaaS provides the highest level of consumer control, while SaaS provides very limited administrative control over the implementation.

1.2 Primary Cloud Service Models

1.2.1 Infrastructure-as-a-Service (IaaS)

The book describes IaaS as providing access to **infrastructure-based IT resources**.

The consumer receives a high level of administrative control over **virtualized infrastructure resources**.

In simple words

IaaS provides infrastructure resources as a cloud service.

1.2.2 Platform-as-a-Service (PaaS)

PaaS provides a **pre-defined, ready-to-use environment**.

The environment contains **pre-packaged products and tools** that support application delivery.

In simple words

PaaS provides a ready-made platform/environment for developing and deploying applications.

1.2.3 Software-as-a-Service (SaaS)

SaaS provides a **software program as a shared cloud service**.

The software is made available as a product or generic utility.

In simple words

SaaS provides software as a cloud service that the consumer can use.

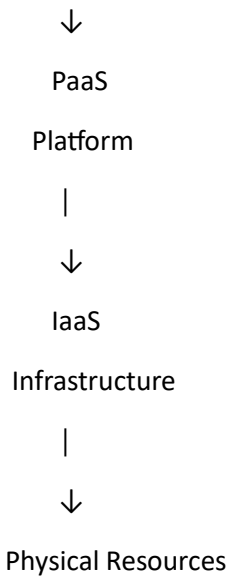
1.3 Relationship Among Service Models

The three delivery models can be understood as layers:

SaaS

Software / Service

|



The book describes the models as a **natural provisioning hierarchy**, in which a PaaS environment can be built upon infrastructure comparable to resources provided by IaaS.

1.3.1 IaaS as Infrastructure Layer

IaaS provides the infrastructure foundation.

The infrastructure includes resources associated with:

- Processing
- Storage
- Networking
- Hosting

The provider provisions and manages the physical processing, storage, networking and hosting resources.

1.3.2 PaaS as Platform Layer

PaaS provides the ready-made platform/environment above the infrastructure.

It provides:

- Pre-packaged products
- Tools
- Application delivery environment

The consumer uses this environment to develop, test and deploy applications/services.

1.3.3 SaaS as Software/Application Layer

SaaS is the software/service layer.

The provider makes the software available as a shared cloud service.

The consumer mainly uses and configures the service.

1.3.4 Increasing Provider Responsibility

As we move from:

IaaS → PaaS → SaaS

more of the underlying environment is managed by the provider.

IaaS



More consumer management

PaaS



More provider management

SaaS



Still more provider management

The reference-based course material explicitly states that more underlying management is handled by the provider as we move from IaaS to PaaS to SaaS.

2. Infrastructure-as-a-Service (IaaS)

2.1 Definition and Concept

Definition

Infrastructure-as-a-Service (IaaS) provides access to infrastructure-based IT resources, with the cloud consumer receiving a high level of administrative control over virtualized infrastructure resources.

Key idea

IaaS



Infrastructure-based IT resources



Virtualized infrastructure



High consumer control

2.1.1 Infrastructure-Based IT Resources

IaaS provides infrastructure resources required to support IT workloads.

Important resources include:

- Processing
- Storage
- Networking
- Hosting

These resources form the underlying infrastructure required by applications and services.

2.1.2 Virtualized Infrastructure

The consumer receives access to **virtualized infrastructure resources**.

The simplified relationship is:

Cloud Consumer



Virtualized Infrastructure



Physical Infrastructure

The provider manages the underlying physical infrastructure, while the consumer receives administrative control over the virtualized infrastructure.

2.2 IaaS Resources

2.2.1 Processing

Processing resources provide computing capability required for executing workloads and applications.

2.2.2 Storage

Storage resources provide the capacity required for storing data.

2.2.3 Networking

Networking resources provide communication between computing resources, storage and other systems.

2.2.4 Hosting/Infrastructure Resources

Hosting resources provide the infrastructure environment required to run workloads.

The provider provisions and manages the physical processing, storage, networking and hosting resources.

2.3 IaaS Responsibilities

2.3.1 Cloud Provider Responsibilities

The IaaS provider:

- Provisions physical processing
- Manages physical processing
- Provisions physical storage
- Manages physical storage
- Provides networking
- Manages hosting/infrastructure

2.3.2 Cloud Consumer Responsibilities

The IaaS consumer:

- Configures infrastructure
- Installs required software
- Manages required software
- Monitors usage

These activities are identified in the reference-based material.

Responsibility view

IaaS Provider

IaaS Consumer

Physical processing Infrastructure configuration

Physical storage Software installation

IaaS Provider	IaaS Consumer
Networking	Software management
Hosting	Usage monitoring

2.4 Level of Consumer Control

IaaS gives the consumer a **high level of administrative control over virtualized infrastructure resources**.

Therefore:

More control also means more responsibility for the consumer.

IaaS

Consumer



More infrastructure control



More management responsibility

2.4.1 Consumer Configuration and Management

The consumer can configure the infrastructure according to application requirements.

Typical activities include:

- Configuring infrastructure
 - Installing software
 - Managing software
 - Monitoring usage
-

2.5 IaaS as a Service Layer

2.5.1 IaaS and Physical Infrastructure

The relationship can be represented as:

IaaS



Virtualized Infrastructure



Physical Resources

IaaS therefore provides an infrastructure layer above the physical resources.

2.5.2 IaaS as Foundation for Higher-Level Services

IaaS can support higher-level service models.

For example:

PaaS



IaaS Resources



Physical Resources

A platform needs underlying processing, storage, networking and hosting resources. The book's combining-model discussion describes this natural provisioning hierarchy.

3. Platform-as-a-Service (PaaS)

3.1 Definition and Concept

Definition

PaaS provides a pre-defined, ready-to-use environment containing pre-packaged products and tools that support application delivery.

Key idea

PaaS



Ready-made environment



Application development



Application deployment

3.1.1 Pre-Defined and Ready-to-Use Environment

The PaaS environment is already prepared by the provider.

The consumer does not have to build the entire underlying platform environment.

This allows the consumer to focus on application-related activities.

3.1.2 Application Delivery Environment

PaaS provides an environment that supports application delivery.

The consumer can:

1. Develop applications
2. Test applications
3. Deploy applications
4. Manage applications/services

These activities are identified in the course material based on the book.

3.2 PaaS Environment

3.2.1 Pre-Packaged Products and Tools

The PaaS environment contains **pre-packaged products and tools** that support application delivery.

This is one of the important characteristics distinguishing PaaS from IaaS.

3.2.2 Application Development

The consumer uses the PaaS environment for developing applications.

The consumer can concentrate on application functionality instead of managing the underlying infrastructure.

3.2.3 Application Deployment

After development and testing, the application can be deployed in the PaaS environment.

Development



Testing



Deployment

3.3 PaaS Responsibilities

3.3.1 Cloud Provider Responsibilities

The provider:

- Pre-configures the platform
- Provides the ready-made environment
- Provisions underlying resources
- Manages the platform
- Manages underlying infrastructure

The provider-side responsibilities include pre-configuring the platform and provisioning underlying resources.

3.3.2 Cloud Consumer Responsibilities

The consumer:

- Develops applications
 - Tests applications
 - Deploys applications
 - Manages applications/services
-

3.4 Level of Consumer Control

Compared with IaaS, the consumer has **less control over the underlying resources** in PaaS.

Why?

Because the provider manages the platform and underlying infrastructure.

IaaS → More infrastructure control

PaaS → Less infrastructure control

The reference-based material explicitly identifies this difference.

3.4.1 Reduced Infrastructure Management

The main advantage is that the consumer does not have to manage the underlying infrastructure to the same extent as in IaaS.

3.4.2 Focus on Application Development and Delivery

PaaS allows the consumer to concentrate on:

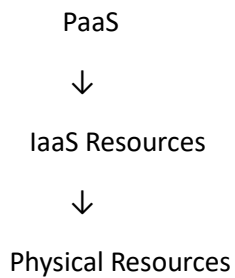
- Application development
- Application testing

- Application deployment
 - Application/service management
-

3.5 Relationship Between PaaS and IaaS

The book describes a **natural provisioning hierarchy**.

A PaaS environment can be built upon infrastructure comparable to resources provided by IaaS.



Why?

The platform requires underlying:

- Processing
- Storage
- Networking
- Hosting

resources.

Therefore:

IaaS can provide the infrastructure foundation for a PaaS environment.

4. Software-as-a-Service (SaaS)

4.1 Definition and Concept

Definition

SaaS provides a software program as a shared cloud service. The software is made available as a product or generic utility.

Key idea

SaaS



Software service



Consumer uses/configures it

4.1.1 Software as a Shared Cloud Service

The software is provided as a cloud service.

The consumer does not need to manage the underlying infrastructure and platform.

4.1.2 Software as a Product or Utility

The software can be consumed as a service rather than being managed as an underlying IT environment by the consumer.

4.2 SaaS Responsibilities

4.2.1 Cloud Provider Responsibilities

The SaaS provider:

- Implements the service
- Manages the service
- Maintains the service
- Monitors the service

These provider activities are identified in the reference-based material.

4.2.2 Cloud Consumer Responsibilities

The consumer mainly:

- Uses the cloud service
 - Configures the service where permitted
-

4.3 Level of Consumer Control

SaaS provides **very limited administrative control over the implementation.**

The consumer mainly interacts with the software service.

IaaS → High control



PaaS → Less underlying control



SaaS → Very limited implementation control

4.3.1 Consumer Primarily Uses/Configures the Service

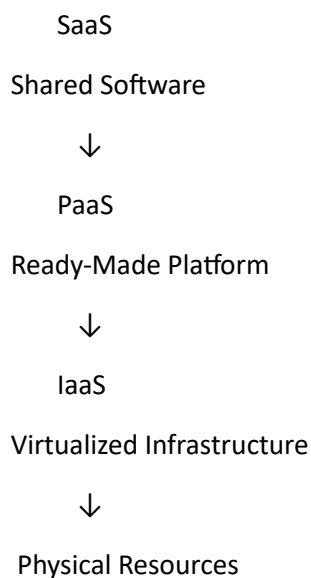
The consumer does not normally manage:

- Physical infrastructure
- Virtualized infrastructure
- Platform implementation
- Software implementation

Instead, the consumer uses and configures the available service.

4.4 Relationship Between SaaS, PaaS and IaaS

The three models can be combined.



The book specifically discusses **combining IaaS + PaaS + SaaS** as part of cloud delivery models.

The course material summarizes the relationship as:

- IaaS provides underlying infrastructure.
 - PaaS provides the ready-made environment.
 - SaaS can be built and offered as a shared cloud service.
-

5. Function as a Service (FaaS)

Syllabus Supplementary Topic

Unit-II syllabus explicitly includes **Function as a Service (FaaS)**.

However, it must be kept separate from the book's core delivery-model framework:

Book's core delivery models: IaaS, PaaS and SaaS.

FaaS: supplementary topic added to satisfy the university syllabus.

5.1 Definition of FaaS

Function as a Service (FaaS) focuses on executing individual application functions in response to events or requests.

The developer supplies the function code, while the cloud platform handles the execution environment and infrastructure.

Simple meaning

FaaS = Execute a particular function when an event or request occurs.

5.1.1 Individual Function Execution

Instead of managing an entire application server, the developer focuses on an individual function.

For example:

Resize an image after a user uploads it.

5.2 Working of FaaS

The basic working is:

Function Code



Cloud Function Service



Event / Request



Function Execution



Result

Step 1 – Function Code

The developer writes the required function.

Step 2 – Function Deployment

The function code is deployed to the cloud provider's function service/code environment.

Step 3 – Event/Request

An event or request occurs.

Step 4 – Function Execution

The cloud platform executes the function.

Step 5 – Result

The function completes the required operation.

5.3 Event-Driven Execution

FaaS is commonly associated with **event-driven execution**.

Event



Function Triggered



Function Executes



Result

5.3.1 Events Triggering Functions

For example:

File Uploaded



Upload Event



Function Triggered



Image Processing

5.3.2 On-Demand Execution

The function executes when it is required by an event or request.

This reduces the need for the developer to directly manage the underlying infrastructure.

5.4 Serverless Concept

FaaS is commonly associated with **serverless computing**.

Important clarification

Serverless does not mean that servers do not exist.

Servers and infrastructure still exist.

The meaning is:

The cloud provider manages the execution infrastructure, while the developer focuses on the function code.

5.4.1 Reduced Server-Management Responsibility

The developer primarily concentrates on:

Function Logic



Function Code

The cloud platform handles the execution environment and infrastructure.

5.5 FaaS Example

File Upload → Image Processing

Suppose a user uploads an image to an application.

The requirement is to automatically resize the image.

User uploads image



File-upload event



FaaS function triggered



Image resizing



Processed image

The function performs only the required image-processing task.

The course material uses this same simple example: a file upload triggers a function that resizes the image.

6. Comparison of Cloud Service Models

6.1 IaaS vs PaaS vs SaaS

The important comparison dimensions are:

- What is provided
- Consumer control
- Consumer responsibility
- Provider responsibility
- Typical use
- Level of abstraction

The reference-based material identifies **consumer control, responsibilities and typical use** as important comparison dimensions.

Comparison Table

Basis	IaaS	PaaS	SaaS
Full form	Infrastructure-as-a-Service	Platform-as-a-Service	Software-as-a-Service
What is provided?	Infrastructure-based IT resources	Pre-defined, ready-to-use environment	Software program as a shared cloud service
Consumer control	High	Lower than IaaS	Very limited
Consumer focus	Infrastructure/software configuration	Application development and delivery	Using/configuring service
Provider focus	Physical infrastructure	Platform and underlying resources	Software service and underlying environment
Typical purpose	Obtain infrastructure resources	Develop/deploy applications	Use software service
Abstraction	Lower	Higher	Highest

6.2 IaaS vs PaaS

Basis	IaaS	PaaS
Provides	Infrastructure-based IT resources	Ready-to-use environment
Consumer control	High	Lower

Basis	IaaS	PaaS
Consumer focus	Infrastructure and software	Application development/delivery
Provider manages	Physical infrastructure	Platform + underlying resources
Main purpose	Infrastructure provision	Application delivery

Main difference

IaaS gives the consumer infrastructure; PaaS gives the consumer a ready-made platform/environment.

6.3 PaaS vs SaaS

Basis	PaaS	SaaS
Provides	Platform/environment	Software service
Consumer activity	Develops, tests and deploys applications	Uses/configures service
Consumer control	Higher	Very limited
Provider responsibility	Platform + underlying resources	Software service + underlying environment
Main purpose	Application delivery	Software consumption

Main difference

PaaS is mainly for application development and delivery, while SaaS is mainly for using a ready-made software service.

6.4 SaaS vs FaaS

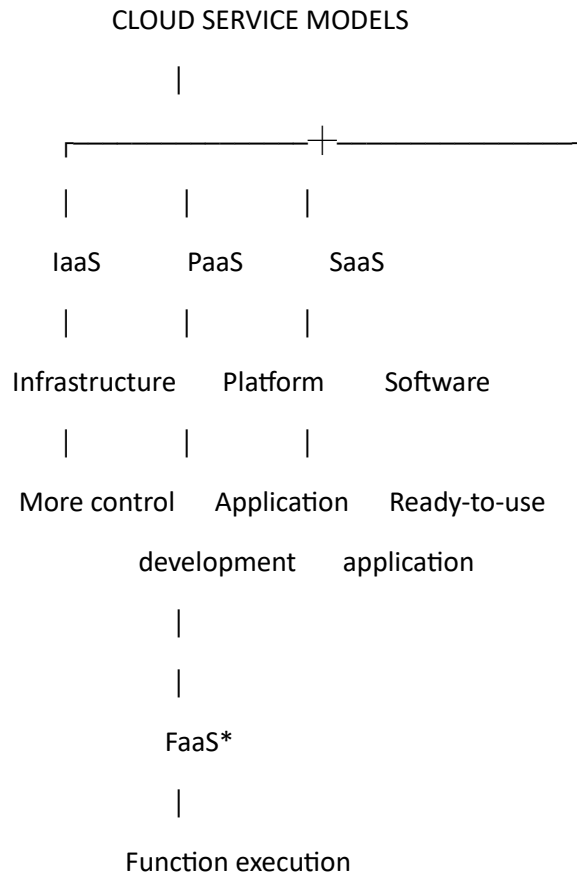
This comparison is syllabus-oriented supplementary material.

SaaS	FaaS
Software service	Individual function execution
Consumer uses/configures service	Developer supplies function code
Application/service-oriented	Function/event-oriented
Provider manages software service	Cloud platform manages execution environment
Generally continuously available as a service	Function executes when triggered

Main difference

SaaS provides a software service; FaaS executes individual functions in response to events or requests.

6.5 Overall Comparison



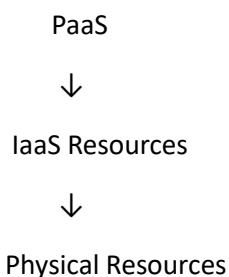
FaaS* = syllabus-required supplementary topic.

7. Combining Cloud Delivery Models

This section is particularly important because the **book itself has a section on combining cloud delivery models**.

7.1 Combining IaaS + PaaS

A PaaS environment can be built upon infrastructure comparable to resources provided through IaaS.



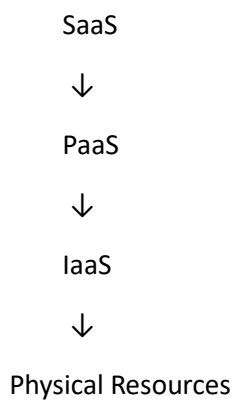
The reason is that the PaaS environment requires underlying:

- Processing
- Storage
- Networking
- Hosting

resources.

7.2 Combining IaaS + PaaS + SaaS

The three base delivery models can be combined:



The roles are:

IaaS → underlying infrastructure

PaaS → ready-made environment

SaaS → shared software service

This layered relationship explains why these models are related rather than isolated.

8. Unit-II Summary

Model Remember This

IaaS Infrastructure-based IT resources

PaaS Pre-defined, ready-to-use environment

SaaS Software program as a shared cloud service

FaaS Individual function execution on events/requests

Control

IaaS → Highest control



PaaS → Less underlying control



SaaS → Very limited implementation control

Main purpose

IaaS → Infrastructure

PaaS → Application development & delivery

SaaS → Software usage

FaaS → Function execution

Provider management

IaaS → PaaS → SaaS



More underlying management by provider

The book-based material confirms this general progression while noting that the exact division of responsibilities depends on the service arrangement.

9. Question bank

Q1. Explain cloud delivery models. Describe IaaS, PaaS and SaaS.

Answer structure:

1. Definition of cloud delivery model
2. Cloud provider and cloud consumer
3. IaaS
4. PaaS
5. SaaS
6. Consumer control
7. Provider responsibility
8. Diagram

Q2. Explain Infrastructure-as-a-Service (IaaS).

Include:

- Definition
- Infrastructure-based IT resources

- Virtualized infrastructure
 - Processing
 - Storage
 - Networking
 - Hosting
 - Provider responsibilities
 - Consumer responsibilities
 - Level of control
-

Q3. Explain Platform-as-a-Service (PaaS).

Include:

- Definition
 - Pre-defined environment
 - Ready-to-use environment
 - Pre-packaged products and tools
 - Application development
 - Application deployment
 - Provider responsibilities
 - Consumer responsibilities
 - Level of control
-

Q4. Explain Software-as-a-Service (SaaS).

Include:

- Definition
 - Shared cloud service
 - Software as product/utility
 - Provider responsibilities
 - Consumer responsibilities
 - Level of administrative control
-

Q5. Explain the relationship among IaaS, PaaS and SaaS.

Draw:

SaaS



PaaS



IaaS



Physical Resources

Explain **combining cloud delivery models**.

Q6. Explain FaaS and its working with a suitable example.

Include:

- Definition
- Function code
- Event/request
- Function execution
- Result
- Event-driven execution
- Serverless concept
- File-upload/image-processing example

Remember: identify FaaS as the **syllabus supplementary topic**, not as one of the book's three core delivery models.

Q7. Compare IaaS, PaaS and SaaS.

Compare:

- What is provided
- Consumer control
- Consumer responsibility
- Provider responsibility
- Typical purpose
- Level of abstraction

Q8. Explain how IaaS, PaaS and SaaS can be combined.

This is especially important because **combining delivery models is explicitly covered by the reference book.**

10. Important Definitions for Examination

Cloud Delivery Model: A model describing how cloud-based IT resources/services are made available to cloud consumers.

IaaS: A delivery model that provides access to infrastructure-based IT resources.

PaaS: A delivery model that provides a pre-defined, ready-to-use environment containing pre-packaged products and tools that support application delivery.

SaaS: A delivery model that provides a software program as a shared cloud service.

FaaS: A supplementary cloud-computing approach in which individual functions are executed in response to events or requests.

Cloud Provider: The party that provides cloud-based IT resources/services.

Cloud Consumer: The party that consumes cloud-based IT resources/services.

Virtualized Infrastructure: Infrastructure resources made available to consumers in virtualized form.

Ready-to-Use Environment: A pre-defined environment containing the products and tools needed to support application delivery.

Event-Driven Execution: Execution of a function in response to an event or request.
